

Optional Task

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```
[1]  
✓ 3s  
import pandas as pd  
import numpy as np  
  
from sklearn.model_selection import train_test_split  
from sklearn.preprocessing import StandardScaler  
from sklearn.compose import ColumnTransformer  
from sklearn.pipeline import Pipeline  
from sklearn.impute import SimpleImputer  
from sklearn.linear_model import LinearRegression  
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score  
import joblib
```

[2]

✓ Os



```
df = pd.read_csv("housing.csv")
```

```
print(df.head())
```

▼

```
... longitude latitude housing_median_age total_rooms total_bedrooms \
0 -122.23 37.88 41.0 880.0 129.0
1 -122.22 37.86 21.0 7099.0 1106.0
2 -122.24 37.85 52.0 1467.0 190.0
3 -122.25 37.85 52.0 1274.0 235.0
4 -122.25 37.85 52.0 1627.0 280.0

population households median_income median_house_value ocean_proximity
0 322.0 126.0 8.3252 452600.0 NEAR BAY
1 2401.0 1138.0 8.3014 358500.0 NEAR BAY
2 496.0 177.0 7.2574 352100.0 NEAR BAY
3 558.0 219.0 5.6431 341300.0 NEAR BAY
4 565.0 259.0 3.8462 342200.0 NEAR BAY
```

[3]

✓ 0s

```
X = df.drop("median_house_value", axis=1)
y = df["median_house_value"]
```

[4]

✓ 0s

```
num_cols = X.select_dtypes(include=['int64', 'float64']).columns  
cat_cols = X.select_dtypes(include=['object']).columns
```

```
print("Numerical Columns:")  
print(num_cols)
```

```
print("\nCategorical Columns:")  
print(cat_cols)
```

▼

```
Numerical Columns:  
Index(['longitude', 'latitude', 'housing_median_age', 'total_rooms',  
       'total_bedrooms', 'population', 'households', 'median_income'],  
      dtype='object')
```

```
Categorical Columns:  
Index(['ocean_proximity'], dtype='object')
```

[5]

✓ Os

```
numeric_transformer = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])

categorical_transformer = Pipeline([
    ("imputer", SimpleImputer(strategy="most_frequent"))
])

preprocessor = ColumnTransformer([
    ("num", numeric_transformer, num_cols),
    ("cat", categorical_transformer, cat_cols)
])
```

[6]

✓ 0s

```
X = pd.get_dummies(X, drop_first=True)
```

[7]

✓ 0s

```
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.2,  
    random_state=42  
)
```

[8]

✓ 0s

```
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
```

```
X_test = scaler.transform(X_test)
```


[9]

✓ 0s

```
model = LinearRegression()  
  
model.fit(X_train, y_train)
```



LinearRegression

1

2

LinearRegression()

[11]

✓ 0s



```
joblib.dump(model, "housing_model.pkl")  
joblib.dump(scaler, "scaler.pkl")  
joblib.dump(X.columns.tolist(), "columns.pkl")
```



... ['columns.pkl']